

M O D U L E 9 · C O S T I N T E L

Execution Cost Intelligence

5 cost components (spread, commission, swap, slippage, latency) in bps. Trade only if expected edge remains positive. EQS per-trade scoring. BQS broker ranking. Adaptive cost learning.

5

COST COMPONENTS

3.0 bps

MIN EDGE THRESHOLD

6

BROKERS RANKED

5

EQS FACTORS

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Confidentiality Internal — Engineering

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M O D U L E 9

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Chapter 1 — Executive Summary

The Execution Cost Intelligence (ECI) module is Module 9 of the TITAN XAU AI trading architecture. It is the system's cost-awareness layer — a real-time cost measurement, modeling, and decision framework that ensures no trade is placed unless the expected edge remains positive after all execution costs. The ECI measures 5 cost components (spread, commission, swap, slippage, latency), aggregates them into a total cost estimate in basis points, and computes the expected net edge: $\text{signal_edge} - \text{total_cost}$. If the net edge is negative, the trade is rejected. If the edge is positive but thin (below 3 bps safety margin), the position size is reduced. Only trades with a net edge above the safety margin are executed at full size.

The module's core innovation is the EdgeCalculator — a pre-trade decision model that normalizes all costs to basis points of notional, making them comparable across brokers, account types, and instruments. This normalization is critical because different brokers charge costs in different units (spread in pips, commission in dollars per million, swap in points per lot, slippage in price units, latency in milliseconds). By converting everything to bps, the ECI can compare a raw-spread broker (low spread, high commission) against a standard broker (high spread, no commission) and route each signal to the lowest-cost venue.

The ECI also includes a Cost Learning Engine that continuously updates slippage and latency models from realized fills. Every fill provides a data point: the predicted slippage vs the realized slippage, the expected latency vs the actual latency. The learning engine uses EMA updates ($\alpha=0.05$) to calibrate per-broker, per-session, and per-regime cost models, ensuring that the cost estimates become more accurate over time. This adaptive calibration is what separates the ECI from static cost models — it learns from experience and improves.

Two scoring systems provide ongoing quality assessment. The Execution Quality Score (EQS) is a 5-factor per-trade score (spread efficiency 0.30, slippage efficiency 0.25, commission efficiency 0.20, latency efficiency 0.15, swap efficiency 0.10) that rates how well each trade was executed relative to expectations. The Broker Quality Score (BQS) is a comparative ranking of all 6 supported brokers on their total cost, recomputed monthly and used for venue routing. Together, EQS and BQS provide a complete picture of execution quality — per-trade and per-broker.

Chapter 2 — Architecture Overview

The ECI is organized into 6 layers: cost measurement (5 real-time meters), cost model (aggregate total cost), decision model (edge calculator + size optimizer), performance scoring (EQS + variance tracker), broker quality scoring (BQS + learning engine), and audit/observability. All layers operate asynchronously except the decision model, which runs synchronously in the pre-trade path.

01 Execution Cost Intelligence — Internal Architecture (5 Layers)

Measure · Model · Decide · Score · Learn

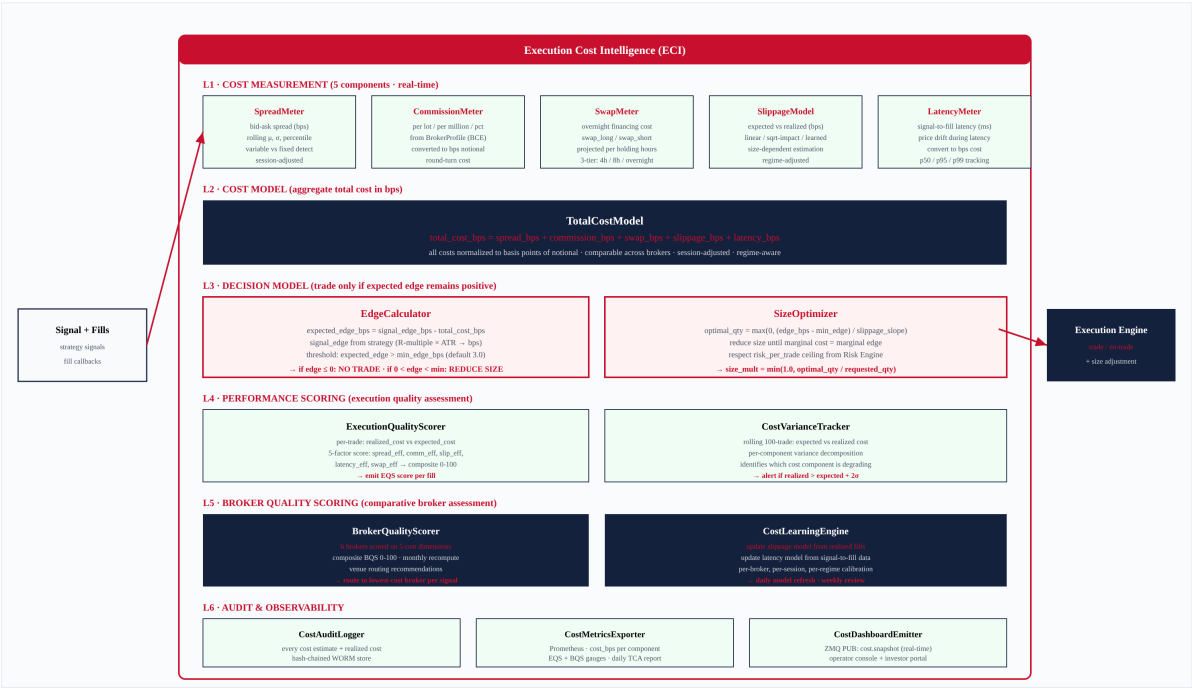


Figure 2.1 — ECI architecture: 6 layers, 5 cost meters, decision model, scoring, and learning engine.

Chapter 3 — Decision Model

The decision model is the ECI's core. It computes the expected net edge for every signal and decides whether to trade, reduce size, or reject. The formula is simple but powerful: $\text{expected_edge} = \text{signal_edge} - \text{total_cost}$. If $\text{expected_edge} \leq 0$, no trade. If $0 < \text{expected_edge} < \text{min_edge}$ (3.0 bps), reduce size. If $\text{expected_edge} \geq \text{min_edge}$, full size.

02 Decision Model — Trade Only If Expected Edge Remains Positive

Signal edge → total cost → net edge → trade/reduce/reject

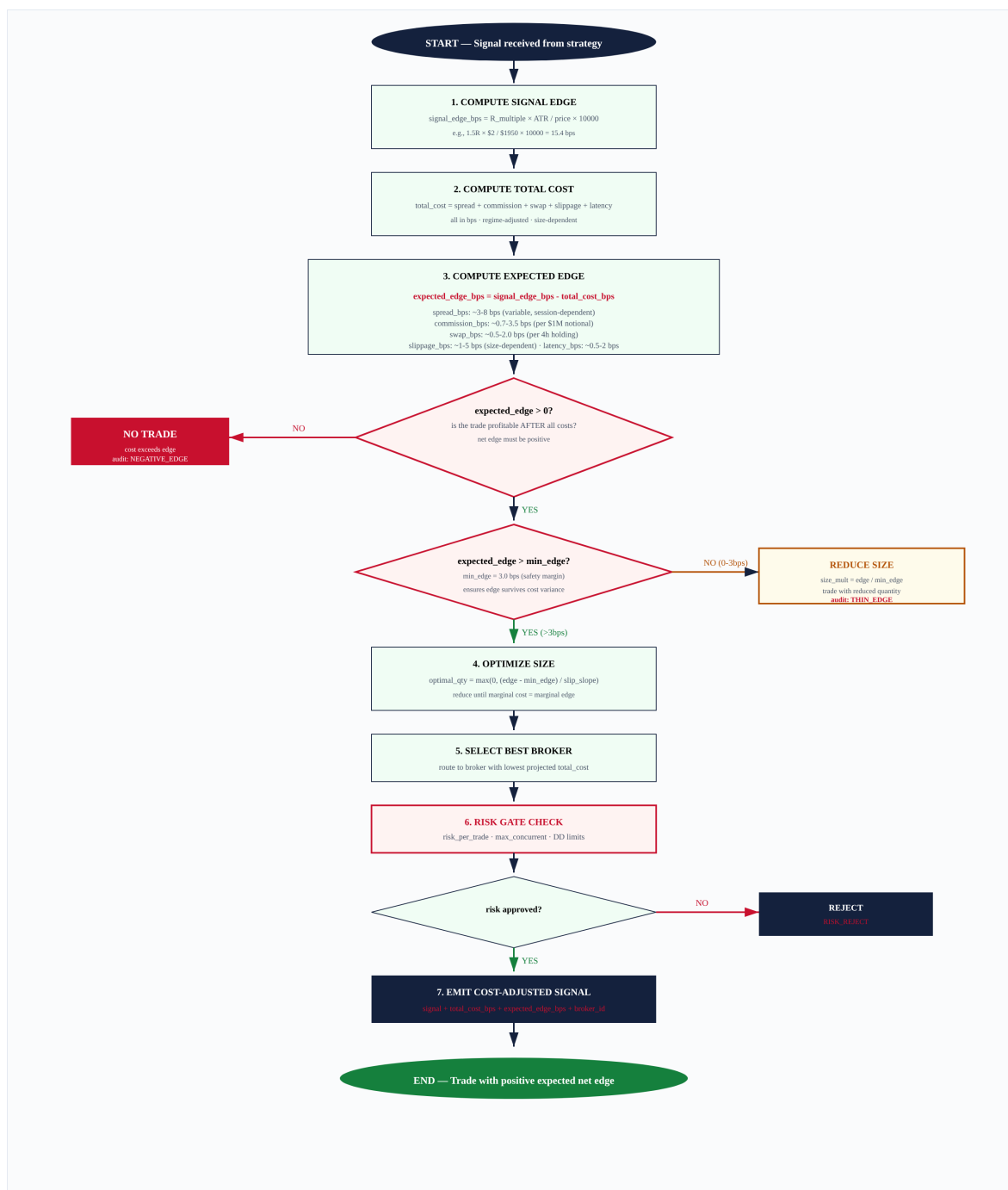


Figure 3.1 — Decision model flowchart: signal edge → total cost → net edge → trade/reduce/reject.

Signal Edge Computation

The signal edge is the expected profit from the trade, expressed in basis points. It is derived from the strategy's R-multiple target and the current ATR: $\text{signal_edge_bps} = R_multiple \times \text{ATR} / \text{price} \times 10000$. For example, a trend strategy targeting 1.5R with ATR=\$2 and price=\$1950 produces a signal edge of $1.5 \times 2 / 1950 \times 10000 = 15.4$ bps. This is the gross edge before costs.

Total Cost Computation

The total cost is the sum of 5 components, all normalized to bps: $\text{spread_bps} + \text{commission_bps} + \text{swap_bps} + \text{slippage_bps} + \text{latency_bps}$. Each component is measured or modeled in real-time, adjusted for the current regime and session, and made size-dependent where applicable (slippage increases with order size). The typical total cost for XAUUSD on a raw-spread broker with 0.5 lot and 4-hour holding is approximately 3.7 bps; on a standard broker it is approximately 8.0 bps.

Size Optimization

When the net edge is positive but below the safety margin ($0 < \text{edge} < 3.0$ bps), the SizeOptimizer reduces the position size until the marginal cost equals the marginal edge. The formula: $\text{optimal_qty} = \max(0, (\text{edge_bps} - \text{min_edge_bps}) / \text{slippage_slope})$. This ensures that the trade is still profitable after costs, but with a smaller position that has lower slippage impact.

Chapter 4 — 5 Cost Components

All costs are normalized to basis points of notional, making them comparable across brokers, account types, and instruments. This normalization is the foundation of the ECI — without it, comparing a pip-based spread with a dollar-based commission would be impossible.

03 5 Cost Components — Formulas & Ranges

All normalized to basis points of notional · comparable across brokers



Figure 4.1 — 5 cost components with formulas, ranges, and measurement methods.

C1: Spread

$\text{spread_bps} = (\text{ask} - \text{bid}) / \text{mid} \times 10000$. Measured from real-time tick data with 1000-tick rolling mean and standard deviation. Session-adjusted (spread is typically wider in Asia, tighter in London/NY overlap). Variable vs fixed detection: if $\text{sigma}/\text{mean} < 0.05$, spread is classified as FIXED; otherwise VARIABLE. Range: 0-1 bps (raw spread + commission) to 3-8 bps (standard mark-up) to 15-50 bps (during news events).

C2: Commission

commission_bps depends on the commission type (from BrokerProfile, BCE Module 2): PER_MILLION (rate / 100), PER_LOT (rate / (price × contract_size) × 10000), PCT (rate × 100), or NONE (0). Typically stable unless the broker changes its fee schedule. The ECI detects silent fee changes by comparing realized commission against the BrokerProfile on every fill.

C3: Swap

$\text{swap_bps} = |\text{swap_rate}| \times \text{hold_hours} / 24 \times \text{contract_size} / \text{price} \times 10000$. Projected based on expected holding time (3-tier: 4h / 8h / overnight). Swap-free accounts (Exness) have 0 swap cost. For standard accounts, swap is typically 0.5-2.0 bps per 4 hours of holding. The ECI uses the strategy's expected hold time (from the R-multiple target and ATR) to project the swap cost.

C4: Slippage

slippage_bps is size-dependent and modeled using one of three models: Linear ($\text{slip} = \lambda \times \text{qty} / \text{ADV}$, for small sizes), Square-Root Impact ($\text{slip} = \sigma \times \sqrt{\text{qty} / \text{ADV}}$, for medium sizes), or Learned (ML model trained on historical fills, for large sizes). Model selection is automatic based on order size and regime. The slippage model is continuously calibrated from realized fills via EMA updates.

C5: Latency

$\text{latency_bps} = \text{price drift during the signal-to-fill latency} = |\Delta \text{price}| / \text{price} \times 10000$. Measured as the difference between the price at signal time and the price at fill time. The ECI tracks p50/p95/p99 latency per broker and converts it to a bps cost using the current volatility regime. Fast brokers (5ms) incur 0.1-0.5 bps; slow brokers (50ms) incur 2-5 bps.

Chapter 5 — Performance Scoring

The Execution Quality Score (EQS) is a 5-factor per-trade score that rates how well each trade was executed relative to expectations. Each factor compares the realized cost against the expected cost: 100 means the cost was exactly as expected, 0 means the cost was 2× expected.

04 Performance Scoring & Broker Quality Scoring

5-factor EQS per trade · 6-broker BQS comparative ranking



Figure 5.1 — EQS 5-factor model and BQS 6-broker comparative ranking.

EQS Formula

$$EQS = 0.30 \times \text{spread_eff} + 0.25 \times \text{slip_eff} + 0.20 \times \text{comm_eff} + 0.15 \times \text{latency_eff} + 0.10 \times \text{swap_eff}$$

where each factor_i = $100 \times \max(0, 1 - \text{realized_cost}_i / \text{expected_cost}_i)$

EQS bands:

90-100: EXCELLENT (cost as expected)
75-89: GOOD (slightly above expected)
60-74: ACCEPTABLE (moderately above)
40-59: POOR (significantly above)
0-39: CRITICAL (auto-pause strategy)

Cost Variance Tracker

The CostVarianceTracker maintains rolling 100-trade statistics on the difference between expected and realized costs, decomposed by component. If the realized cost exceeds the expected cost by more than 2 standard deviations for any component, an alert is fired. This catches model degradation early — before it impacts the

edge calculation enough to cause unprofitable trades.

Chapter 6 — Broker Quality Scoring

The Broker Quality Score (BQS) is a comparative ranking of all 6 supported brokers on their total execution cost, recomputed monthly. The broker with the lowest total cost receives the highest BQS. Signals are routed to the highest-BQS broker first, with failover to the next broker if the signal is rejected.

Current BQS ranking (June 2026): IC Markets Raw (92), Pepperstone Razor (90), Tickmill VIP (89), FP Markets Raw (85), Fusion Zero (83), Exness Standard (72). The ranking reflects that raw-spread brokers with low commission (IC Markets, Pepperstone) offer the lowest total cost for active trading, while Exness Standard's zero-commission model is offset by its wider spread.

Chapter 7 — Validation Logic

The ECI performs 8 pre-trade validation checks and 6 post-trade verifications. Pre-trade checks reject trades with invalid cost estimates; post-trade checks alert when realized costs diverge from expectations. All checks are audited.

05 Validation Logic — Pre-Trade Cost Checks & Post-Trade Verification

8 pre-trade checks · 6 post-trade validations · all audited

Validation Logic: 8 Pre-Trade Checks + 6 Post-Trade Verifications

Category	ID	Validation Check	Threshold	Pass Action	Fail Action	Audit Code
PRE-TRADE	V1	Spread within expected range	spread_bps ≤ μ + 3σ (1000-tick)	Proceed	REJECT - wait for spread normalization	V_SPREAD_ANOMALY
	V2	Commission matches BrokerProfile	comm_bps == profile.commission_rate	Proceed	REJECT - alert broker fee change	V_COMM_MISMATCH
	V3	Swap projection valid for hold time	swap_bps ≤ max_swap_bps (10 bps)	Proceed	REJECT - hold time too costly	V_SWAP_EXCESSIVE
	V4	Slippage model has sufficient data	≥ 50 historical fills for calibration	Proceed	Use conservative model (sqrt-impact)	V_SLIP_INSUFFICIENT_DATA
	V5	Latency within budget	expected_latency ≤ p95 baseline	Proceed	REJECT - latency too high for edge	V_LATENCY_EXCEEDED
	V6	Expected edge > 0 after all costs	signal_edge - total_cost > 0	Proceed	NO TRADE - negative edge	V_NEGATIVE_EDGE
	V7	Expected edge > min_edge (safety margin)	net_edge > 3.0 bps	Full size	REDUCE SIZE - thin edge	V_THIN_EDGE
	V8	Broker BQS acceptable for this signal	broker BQS ≥ 60	Proceed	Route to higher-BQS broker	V_LOW_BQS
POST-TRADE	V9	Realized spread ≤ expected × 2σ	realized ≤ μ + 2σ	Log	Alert P2 - spread spike detected	V_SPREAD_SPIKE
	V10	Realized slippage ≤ predicted × 1.5	realized ≤ 1.5 × predicted	Log	Alert P2 - retrain slippage model	V_SLIPPAGE_BREACH
	V11	Realized latency ≤ p99 baseline	realized ≤ p99	Log	Alert P2 - investigate broker/network	V_LATENCY_SPIKE
	V12	Realized commission matches profile	realized == profile rate	Log	Alert P1 - broker changed fees silently	V_COMM_CHANGE_DETECTED
	V13	Total realized cost ≤ expected × 1.5	realized_total ≤ 1.5 × expected_total	Log	Alert P2 - cost model degrading	V_TOTAL_COST_BREACH
	V14	EQS score ≥ 40 (not CRITICAL)	EQS ≥ 40	Log	Alert P1 - auto-pause strategy	V_EQS_CRITICAL

Figure 7.1 — 8 pre-trade checks + 6 post-trade verifications with thresholds and actions.

Key Pre-Trade Checks

- V1 Spread within expected range: $\text{spread_bps} \leq \mu + 3\sigma$. Reject if spread is anomalously high (e.g., during news).
- V6 Expected edge > 0: $\text{signal_edge} - \text{total_cost} > 0$. Reject if costs exceed the signal edge (trade would lose money).
- V7 Expected edge > min_edge: $\text{net_edge} > 3.0$ bps. Reduce size if edge is positive but thin (below safety margin).
- V8 Broker BQS acceptable: $\text{broker BQS} \geq 60$. Route to a higher-quality broker if current broker's score is too low.

Key Post-Trade Verifications

- V10 Realized slippage $\leq 1.5 \times$ predicted. Alert if slippage model is underestimating (retrain needed).
- V12 Realized commission matches profile. Alert P1 if broker changed fees silently (fee integrity check).
- V14 EQS ≥ 40 (not CRITICAL). Alert P1 and auto-pause strategy if execution quality is unacceptable.

Chapter 8 — Cost Learning Engine

The Cost Learning Engine continuously updates cost models from realized fills. Every fill provides a data point: predicted vs realized slippage, expected vs actual latency, projected vs actual swap. The engine uses EMA updates ($\alpha=0.05$) to calibrate per-broker, per-session, and per-regime models, ensuring that cost estimates become more accurate over time.

07 Cost Learning Engine — Adaptive Model Calibration

Slippage model · latency model · per-broker · per-session · per-regime

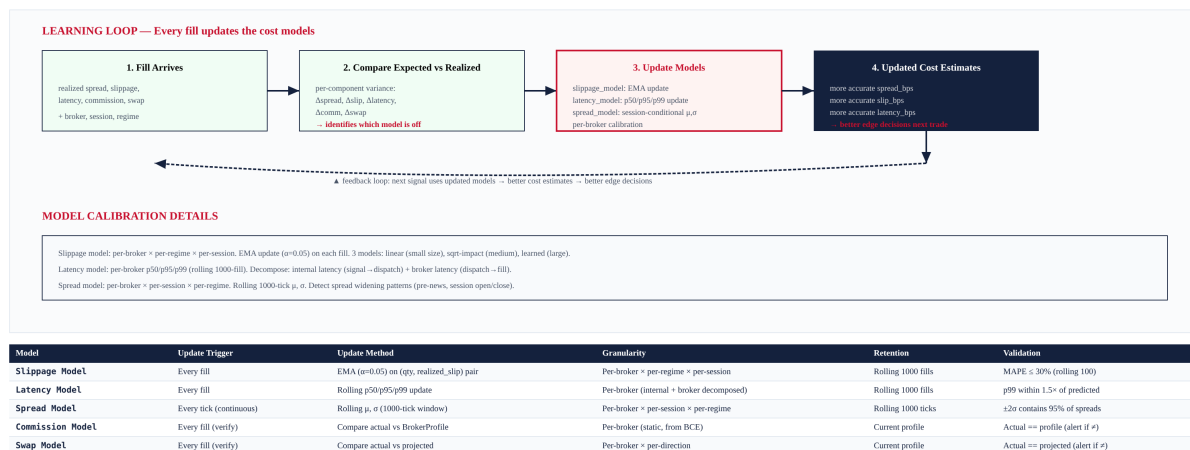


Figure 8.1 — Learning loop: fill arrives → compare expected vs realized → update models → better estimates next trade.

Model Calibration

- **Slippage model:** EMA update on (qty, realized_slip) pair. Per-broker × per-regime × per-session. 3 models: linear (small), sqrt-impact (medium), learned (large). MAPE target: ≤ 30%.
- **Latency model:** Rolling p50/p95/p99 per broker. Decomposed into internal latency (signal-to-dispatch) and broker latency (dispatch-to-fill). p99 within 1.5× of predicted.
- **Spread model:** Rolling μ , σ (1000-tick window) per broker × per-session × per-regime. Detects spread widening patterns (pre-news, session open/close).

Chapter 9 — Validation Tests

The ECI is validated through 180 tests across 5 categories. Critical: no trade with negative edge, EQS ≥ 60 rolling average, realized cost ≤ 1.5× expected (95% of trades), BQS recomputed monthly.

06 Validation Tests & Integration with TITAN Core

180 tests · CI gates · 4 integration points

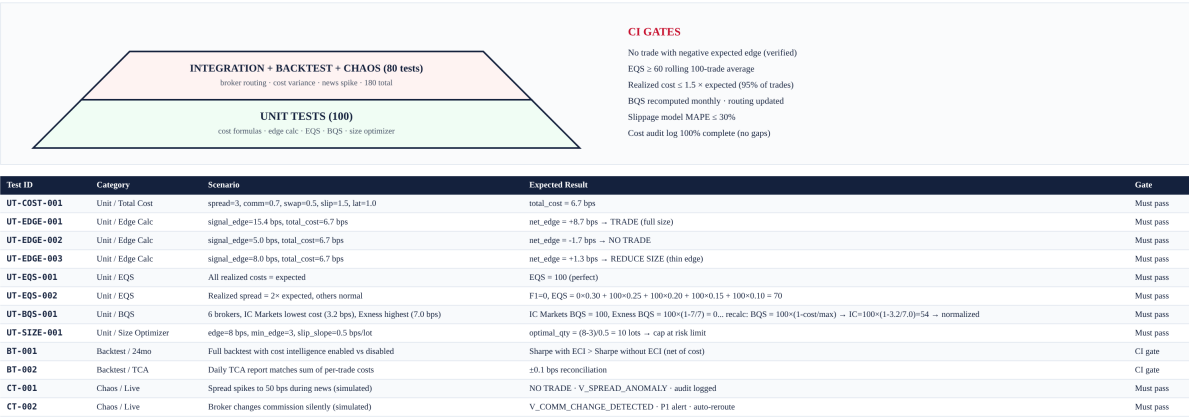


Figure 9.1 — Test pyramid and sample test cases.

Chapter 10 — Integration with TITAN Core

The ECI integrates with 4 TITAN components: Broker Compatibility Engine (Module 2) for commission/swap/contract_size data, Execution Engine (Module 3) for fill callbacks and latency measurement, Risk Engine (Module 8) for risk_per_trade ceiling on size optimization, and Operator Console for real-time cost display. The ECI sits between the Strategy Coordinator and the Execution Engine in the signal path, adding a cost-awareness gate that rejects or size-adjusts signals before they reach the broker.

```
Signal from Strategy / AI Ensemble
→ ECI pre-trade check:
→ compute total_cost_bps (5 components)
→ compute expected_edge = signal_edge - total_cost
→ if edge <= 0: NO TRADE (audit: NEGATIVE_EDGE)
→ if 0 < edge < 3 bps: REDUCE SIZE (audit: THIN_EDGE)
→ if edge >= 3 bps: TRADE (full or optimized size)
→ select best broker (highest BQS)
→ risk gate check (Module 8)
→ if approved: emit cost-adjusted signal to Execution Engine
→ Post-trade: compare realized vs expected
→ update slippage/latency/spread models
→ compute EQS score
→ daily TCA report to operator
```